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Electron Ion Collider Parameter List Summary

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UNDER CONSTRUCTION

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Chapter 1

Introduction

The EIC collides polarized electrons in an Electron Storage Ring (ESR) with ions in a Hadron Storage Ring (HSR), with species that range from polarized protons and deuterons to unpolarized heavy ions. Figure 1.1 illustrates the scope of the EIC project.

Electron injector complex. A new electron injector complex delivers full-energy electron bunches to the ESR. It comprises a high-intensity preinjector (a polarized electron source and a linac), a Beam Accumulator Ring (BAR), and a Rapid Cycling Synchrotron (RCS). Each ESR electron bunch is replaced every few minutes, in order to maintain the required average ESR beam polarization. To do this the electron injector provides a new electron bunch at a rate of at least 1 Hz.

Preinjector. The polarized electron source is a polarized, room temperature photo-cathode electron gun. The subsequent copper S-band linac is installed in a new enclosure, connected to the Beam Accumulator Ring (BAR) and the RCS ring enclosure.

Beam Accumulator Ring. The new BAR ring is installed between the S-band linac and the RCS, with a circumference of about 40 m. It accumulates highly-polarized electron bunches from the preinjector before transportation to the RCS.

Rapid Cycling Synchrotron. The RCS is installed in a new enclosure near RHIC Insertion Region 4 (IR4), with a circumference of 1,421 m. It accelerates highly-polarized electron bunches to the ESR collision energy after extraction from the BAR, using normal conducting magnets and superconducting RF cavities.

Electron Storage Ring. The ESR is a new ring installed in the RHIC tunnel, with a circumference close to that of the HSR. The HSR and ESR revolution frequencies are locked when injection is complete, synchronizing bunch collisions. The ESR is comprised of room temperature magnets, superconducting RF systems, and a vacuum system that can absorb up to 10 MW of synchrotron radiation.

Hadron Storage Ring. The HSR is a modified version of the existing RHIC yellow ring; the RHIC blue ring will be removed. Several HSR modifications and upgrades are necessary to support HSR beam parameters, including removal of DX magnets to enhance the

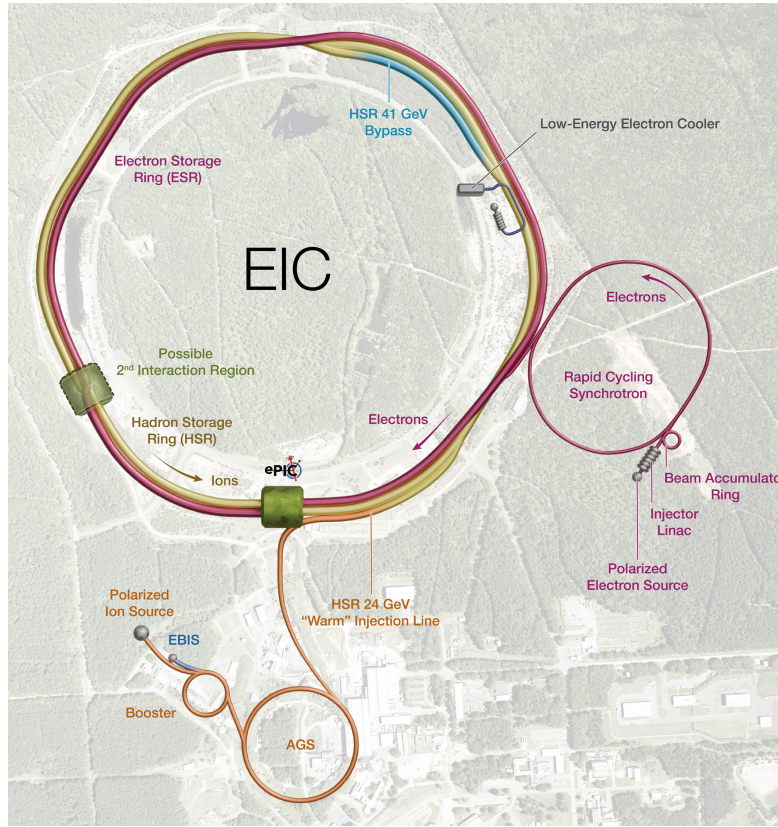


Figure 1.1: The EIC scope starts downstream of the “YD26” dipole in the last section of the AGS-to-RHIC transfer line. For the EIC, the warm transfer line is extended to the IR4 HSR area, where the beam is injected into the cold HSR. Apart from this transfer line extension, the EIC scope does not include any other modifications to the existing Hadron Injector.

maximum hadron energy, addition of a superconducting RF system, and a revised AGS-to-RHIC (AtR) injection beamline. The HSR beampipe vacuum chamber, bellows, and diagnostics are modified and updated. The existing RHIC superconducting hadron spin rotators are upgraded and relocated to support high beam polarization.

Cryogenic systems. The Collider-Accelerator Department (C-AD) upgrades the RHIC Liquid Helium (LHe) central refrigerator plant to improve efficiency, reduce helium loss, and meet the EIC cryogenic load requirements. The EIC project provides 2 K helium to the SRF cavities and the new superconducting magnets near the interaction point, using three new satellite plants that draw 4.6 K liquid helium from the central cryogenic plant.

Collider Parameters. The following two tables list reference EIC collider beam parameters for representative ion species and center of mass collision energies. Full ranges of colliding beam energies are identified in the respective ESR and HSR chapter tables.

The parameters in this document are based on the EIC design evolution from that described in the Conceptual Design Report. Changes since that time are documented in the upcoming Preliminary Design Report.

1.0.1 Proton collisions – 275 GeV

Parameter	Unit	Values		Comment
Center-of-mass energy E_{cm}	GeV	105	141	
Proton energy	GeV	275	275	Total particle energy
Electron energy	GeV	10	18	Total particle energy
Luminosity L	$10^{33}\text{cm}^{-2}\text{s}^{-1}$	10.0	1.54	Total RF power limit reduces luminosity at 18 GeV electron beam energy
Hourglass factor H		0.94	0.91	
No. of bunches M		1160	290	
RMS beam size σ_H	μm	95	119	
σ_V	μm	8.5	11.0	
$K_x \equiv \sigma_H / \sigma_V$		11.1	11.1	
Protons				
Bunch intensity	10^{10}	6.9	19.1	
Beam current	A	1.00	0.69	
Polarization	%	70	70	Longitudinal and transverse
RMS normalized emittance H	μm	3.3	5.2	Flat beams with a size ratio of 11
V	μm	0.30	0.47	
RMS emittance H	nm	11.3	18.0	
V	nm	1.0	1.6	
Collision β^* H	m	0.80	0.80	
V	m	0.072	0.071	
RMS beam divergence H	μrad	119	150	
V	μrad	119	150	
Beam-beam parameter ξ H		0.012	0.003	
V		0.012	0.003	
RMS longitudinal emittance	eV.s	0.036	0.036	
RMS bunch length	mm	60	60	
RMS $\Delta p/p$	10^{-3}	0.68	0.68	
Maximum space charge ΔQ		0.004	0.007	
Piwinski angle	rad	7.9	6.3	
IBS growth time L	h	2.9	2.0	
H	h	2.0	2.0	
V	h	67	82	
Electrons				
Bunch intensity	10^{10}	17.2	6.2	
Beam current	A	2.5	0.227	
Polarization	%	70	70	Longitudinal only
RMS normalized emittance H	μm	391	845	
V	μm	26	71	
RMS emittance H	nm	20	24	
V	nm	1.3	2.0	
Collision β^* H	m	0.55	0.70	
V	m	0.065	0.057	
RMS beam angle H	μrad	211	202	
V	μrad	152	187	
Beam-beam parameter ξ H		0.072	0.093	
V		0.100	0.100	
RMS bunch length	mm	7	9	
RMS $\Delta p/p$	10^{-3}	0.58	1.09	
Piwinski angle	rad	2.4	2.1	

Table 1.1: Proton collisions – 275 GeV

1.0.2 Proton collisions – 100 GeV

Parameter	Unit	Values		Comment
Center-of-mass energy E_{cm}	GeV	44.7	63.2	
Proton energy	GeV	100	100	Total particle energy
Electron energy	GeV	5	10	Total particle energy
Luminosity L	$10^{33} \text{cm}^{-2} \text{s}^{-1}$	3.68	4.48	
Hourglass factor H		0.88	0.90	
No. of bunches M		1160	1160	
RMS beam size σ_H	μm	125	138	
σ_V	μm	11	12	
$K_x \equiv \sigma_H/\sigma_V$		11.1	11.1	
Protons				
Bunch intensity	10^{10}	4.8	6.9	
Beam current	A	0.69	1.0	
Polarization	%	70	70	Longitudinal and transverse
RMS normalized emittance H	μm	2.7	3.2	
V	μm	0.25	0.29	
RMS emittance H	nm	26	30.0	
V	nm	2.3	2.7	
Collision β^* H	m	0.61	0.63	
V	m	5.5	5.7	
RMS beam divergence H	μrad	206	220	
V	μrad	206	220	
Beam-beam parameter ξ H		0.014	0.012	
V		0.014	0.012	
RMS longitudinal emittance	eV.s	0.021	0.021	
RMS bunch length	mm	70	70	
RMS $\Delta p/p$	10^{-3}	0.97	0.97	
Maximum space charge ΔQ		0.021	0.026	
Piwinski angle	rad	7.0	6.3	
IBS growth time L	h	3.1	2.5	
H	h	2.0	2.0	
V	h	4.0	4.0	
Electrons				
Bunch intensity	10^{10}	17.2	17.2	
Beam current	A	2.5	2.5	
Polarization	%	70	70	Longitudinal only
RMS normalized emittance H	μm	196	391	
V	μm	18	26	
RMS emittance H	nm	20	20	
V	nm	1.8	1.3	
Collision β^* H	m	0.78	0.61	
V	m	0.12	0.071	
RMS beam angle H	μrad	160	145	
V	μrad	160	105	
Beam-beam parameter ξ H		0.100	0.072	
V		0.100	0.100	
RMS bunch length	mm	7.0	7.0	
RMS $\Delta p/p$	10^{-3}	0.68	0.58	
Piwinski angle	rad	2.0	1.8	

Table 1.2: Proton collisions – 100 GeV

1.0.3 Proton collisions – 41 GeV

Parameter	Unit	Value	Comment
Center-of-mass energy E_{cm}	GeV	28.6	
Proton energy	GeV	41	Total particle energy
Electron energy	GeV	5	Total particle energy
Luminosity L	$10^{33}\text{cm}^{-2}\text{s}^{-1}$	0.44	
Hourglass factor H		0.93	
No. of bunches M		1160	
RMS beam size σ_H	μm	198	
σ_V	μm	27	
$K_x \equiv \sigma_H / \sigma_V$		7.3	
Protons			
Bunch intensity	10^{10}	2.6	
Beam current	A	0.38	
Polarization	%	70	Longitudinal and transverse
RMS normalized emittance H	μm	1.9	
V	μm	0.45	
RMS emittance H	nm	44	
V	nm	10	
Collision $\beta^* H$	m	0.90	
V	m	0.071	
RMS beam divergence H	μrad	380	
V	μrad	220	
Beam-beam parameter ξH		0.015	
V		0.009	
RMS longitudinal emittance	eV.s	0.011	
RMS bunch length	mm	75	
RMS $\Delta p/p$	10^{-3}	1.03	
Maximum space charge ΔQ		0.05	
Piwinski angle	rad	4.2	
IBS growth time L	h	3.8	
H	h	3.4	
V	h	2.1	
Electrons			
Bunch intensity	10^{10}	13.3	
Beam current	A	1.93	
Polarization	%	70	Longitudinal only
RMS normalized emittance H	μm	196	
V	μm	34	
RMS emittance H	nm	20	
V	nm	3.5	
Collision $\beta^* H$	m	1.96	
V	m	0.21	
RMS beam angle H	μrad	101	
V	μrad	129	
Beam-beam parameter ξH		0.053	
V		0.042	
RMS bunch length	mm	7.0	
RMS $\Delta p/p$	10^{-3}	0.68	
Piwinski angle	rad	1.1	

Table 1.3: Proton collisions – 41 GeV.

1.0.4 Helium-3 collisions – 183 GeV

Parameter	Unit	Values		Comment
		He-3	Electrons	
Center-of-mass energy E_{CM}	GeV		85.6	
Beam energy	GeV	183.0	10.0	
Bunch intensity	10^{10}	3.000	17.203	
Number of bunches			1160	
Beam current	A	0.87	2.500	
RMS normalized emittance, h/v	μm	2.0/0.30	391/27.6	
RMS emittance, h/v	nm	10.3/1.5	20.0/1.4	
emittance _y /emittance _x		0.149	0.070	
beta,h/v	cm	80/7.0	41/7.6	
IP beamsize, h/v	um	91/10.4	91/10.4	
$K=\sigma_y/\sigma_x$			0.114	
IP RMS angular spread, h/v	μrad	114/148	220/136	
beam-beam parameter, h/v		0.012/0.010	0.062/0.100	
longitudinal bunch area, evs		0.80		
RMS bunch length	cm	6.0	2.0	
RMS energy spread	10^{-4}	6.6	5.8	
Max space charge		0.006	neglig.	
Piwinski angle	rad	8.2	2.7	
Longitudinal IBS time	h	1.56		
Transverse IBS time	h	0.95		
Luminosity	10^{33}		10.90	

Table 1.4: Helium-3 collisions – 183 GeV

Chapter 2

Electron Injector

The **Electron Injector** complex generates, accumulates, accelerates, and injects up to two 28 nC polarized electron bunches per second into the Electron Storage Ring (ESR). The **Preinjector** produces ~ 1 nC polarized bunches at 30 Hz, and accelerates them to 750 MeV. The **Beam Accumulator Ring** accumulates and equilibrates high-intensity bunches from the preinjector. The **Rapid Cycling Synchrotron** accepts polarized electron bunches from the BAR or Preinjector, and accelerates and delivers them to ESR injection.

2.0.1 Preinjector

Parameter	Unit	Value
General		
Beam extraction energy	MeV	750
Beam repetition rate	Hz	30
Energy spread at end of linac	%	0.25
Normalized RMS horizontal emittance	μm	60
Normalized RMS vertical emittance	μm	40
Bunch charge	nC	1
Polarization direction		Vertical
Polarization	%	≥ 86
RMS bunch length	mm	0.75
Electron Gun		
Gun voltage	kV	310
Bunch length	ps	300
Laser parameters		
Laser pulse energy	μJ	≤ 6
Wave length tunable range	nm	760 -790
Repetition rate	Hz	30
Polarization	%	90
Polarization switch speed		1 Hz in 30 ms
Pulse duration	ps	200 - 350
Timing jitter (RMS)	ps	< 3
Low Energy RF		
Buncher frequency	MHz	197
Buncher voltage	kV	500
Capture cavities frequency	MHz	1300
Capture cavity 1 voltage	MV	5
Capture cavity 2 voltage	MV	10
Electron Linac		
Number of TW accelerating units		14
Linac frequency	MHz	2856
Linac voltage	MV	60

Table 2.1: Preinjector lattice parameters. Electron polarization is longitudinal for 1.7 m from the gun to the spin rotator, where it is rotated to the vertical direction for the rest of the ESR injection process.

2.0.2 Beam Accumulator Ring

Parameter	Unit	Value
General		
Number of bunches		1
Bunch length	mm	43
Betatron tune, H		3.12
V		1.18
Injection		
Energy	MeV	750
Polarization	%	86
Bunch charge	nC	1
Repetition rate	Hz	30
Horizontal RMS emittance (norm)	μm	~ 30
Vertical RMS emittance (norm)	μm	~ 50
RMS energy spread		0.0045
Extraction		
Energy	MeV	750
Polarization	%	86
Bunch charge	nC	28
Repetition rate	Hz	1
Horizontal RMS emittance (geom)	nm	124
Horizontal RMS emittance (norm)	μm	182
Vertical RMS emittance (geom, 3% coupling)	nm	4
Vertical RMS emittance (norm, 3% coupling)	μm	5.9
RMS energy spread	10^{-3}	0.47
RF		
Frequency	MHz	52.88
Voltage	kV	100
Number of normal conducting cavities		1
Magnet		
Main dipole number		8
Main dipole effective length	m	1.5
Main dipole field	T	1.3099
Quad number		24
Quad effective length	m	0.3
Quad field gradient	T/m	4.6/-2.9/4.7

Table 2.2: Beam Accumulator Ring lattice parameters.

2.0.3 Rapid Cycling Synchrotron

Parameter	Unit	Value	Comment
General			
Repetition rate	Hz	1	
Acceleration time	ms, turns	100, 21000	
Number of bunches		1	
Bunch charge	nC	1–28	Varies with operating scenario
Circumference	m	1421.257	
Betatron tune, H		43.12	
V		56.16	
Max beta functions (x,y)	m	30.48, 35.73	
Max dispersion (x)	m	0.59	
Momentum compaction	10^{-4}	6.48	
Injection			
Energy	MeV	750	
RMS emittance (norm; x,y)	μm	<180	
RMS energy spread	10^{-3}	<0.5	
RMS bunch length	mm, ps	<12, <40	
Extraction			
Energy	GeV	5–18	
Polarization, 5 GeV	%	85	
10 GeV	%	85	
18 GeV	%	85	
RMS emittance (norm; x,y)	μm	196–845, 18–113	
RMS energy spread	10^{-3}	0.58–1.09	
RMS bunch length	mm, ps	7–9, 23–30	
Radiation per electron (18 GeV)	MeV/turn	101.3	
Horizontal damping time (18 GeV)	ms	2.75	
RF			
Frequency	MHz	591.039	
Voltage	MV	130	
Magnet/Vacuum			
Main dipole number		320	
Main dipole effective length	m	1.8	
Main dipole bend angle	mrad	19.635	
Main dipole field (max)	T	0.65	
Quad number		352	
Quad effective length	m	0.6	
Quad strength K_1 (max)	m^{-2}	0.785	
Sext number		640	
Sext effective length	m	0.2	
Sext strength K_2 (max)	m^{-3}	27.5	
Beam pipe ID, thickness	mm	36.32, 0.89	

Table 2.3: Rapid Cycling Synchrotron lattice parameters.

Chapter 3

Electron Storage Ring

The ESR is installed in the existing RHIC tunnel, where it intersects with the HSR in IR6 and IR8. IR6 is equipped with spin rotators, crab cavities, and dedicated low- β magnets, while IR8 contains only a simple cross-over section with the future potential to host a second detector. The circumference ensures that the electron revolution frequency matches that of 133 GeV on-axis proton beams.

The ESR plane is tilted by 200 μrad with respect to the HSR, about an axis through IP6 and IP8. This results in elevation differences of 104 mm and 208 mm at IP4 and IP12, facilitating the accelerator cross-overs there.

Arc magnets are arranged in a FODO cell pattern. Each half-cell contains two 2.73 m long D13 dipoles and a single 0.89 m D2 dipole, in a “super-bend” structure. All three dipoles are powered uniformly at energies of 10 GeV and above, minimizing synchrotron radiation losses. The D2 dipoles are powered with reverse polarity below 10 GeV, increasing the equilibrium emittance to about 20 nm and increasing radiation damping in order to support a large beam-beam tune shift parameter even at low energy. Arc sextupoles are powered to maximize the dynamic aperture, in family structures that depend on the beta-tron phase advance in the arc FODO cells, which in turn is chosen depending on the beam energy.

3.0.1 Top level

Parameter	Unit	Values		
Beam energy	GeV	5	10	18
Average polarization	%	70	70	70
Asymptotic polarization	%	16.0	47.4	33.9
Sokolov-Ternov time constant	min	1124	679	33.4
Bunch replacement time, spin up spin down	min	149	451	10.1
	min	231	98.6	3.75
Maximum circulating current	mA	2500	2500	230
Bunch count		1160	1160	290
Maximum bunch charge	nC	28	28	11
Harmonic number		7560	7560	7560
Natural unnormalized RMS emittance	nm	20	20	30
RMS fractional momentum spread	10^{-3}	0.48	0.54	0.97
RMS bunch length	mm	7	7	9
Filling rate	bunches/s	1	1	1
Filling time	min	20	20	5
Beam stored energy	kJ	162	325	57.4

Table 3.1: Top level Electron Storage Ring parameters. The Sokolov-Ternov **asymptotic polarization** values apply in the absence of swap-out injection. An average electron polarization of 70% is maintained by frequent bunch replacement. The polarization direction is vertical in the arcs, and longitudinal at the collision point. Spin rotators manipulate the turn-by-turn changes between vertical and longitudinal.

3.0.2 Lattice and orbit

Parameter	Unit	Values		
Beam energy	GeV	5	10	18
Circumference	m	3834.0	3834.0	3834.0
Revolution frequency	kHz	78.193	78.193	78.193
Radio frequency	MHz	591	591	591
Harmonic number		7560	7560	7560
Energy loss per turn	MeV	1.15	3.71	40
Momentum compaction factor	10^{-3}	0.60	1.19	0.60
Betatron tune, H		43.08	43.08	51.08
V		35.14	35.14	43.14
RF Voltage	MV	5.0	22.5	56.0
Number of FODO cells per arc		16	16	16
Phase advance per arc FODO cell	deg.	60	60	90
Average arc bending radius	m	380	380	380
Arc dipole bending radius, D13	m	-52.44	262.26	262.26
D2	m	132.37	262.26	262.26
Arc dipole field, D13	T	-0.32	0.12	0.23
D2	T	0.13	0.12	0.23
Arc quadrupole gradient	T/m	3.74	7.22	18.56
Number of sextupole families per arc		6	6	4
Maximum integrated sextupole gradient	T/m	24.5	65.0	111.2

Table 3.2: Electron Storage Ring lattice and orbit parameters

Chapter 4

Hadron Storage Ring

The HSR, based on the Yellow ring, inherits RHIC's versatility to collide a large range of ion species over a wide energy range. It operates with three times more hadron beam current, a larger number of bunches, and a 10% larger maximum beam energy. Copper and amorphous carbon coated beam screens are installed in the vacuum chamber to mitigate resistive wall heating and electron cloud effects. The proton beam circumference is varied by changing the radial orbit, in order to synchronize collisions with the electron beam over an energy range from 100 to 275 GeV. Proton collisions at 41 GeV energy are realized by using an inner arc bypass in the sextant from IP12 to IP2. The warm AtR injection line is considerably extended, bringing hadron beams to IR4 before injection into cold HSR magnets.

4.0.1 Top level

Parameter	Unit	Values	
		Protons	Gold
Injection energy	GeV or GeV/u	23.8	10.0
Top energy	GeV or GeV/u	275	110
Injection polarization	%	72	–
Longitudinal/Transverse average polarization	%	70	–
Betatron tune, H		27.23	28.23
V		26.21	29.21
Bunch count, injection		290	290
top		1160	1160
Ions per bunch at store		69×10^9	0.5×10^9
Beam current	A	1.0	0.57
Beam stored energy	MJ	3.53	1.97

Table 4.1: Top level Hadron Storage Ring parameters. The average proton polarization of 70% during collisions is an assumption.

4.0.2 Layout, lattice, and RF

Parameter	Unit	Value
Layout and lattice		
Magnetic rigidity, injection	Tm	79.3
top	Tm	917.3
Arc dipole, bend radius	m	242.78
top field	T	3.804
current	kA	5.602
Arc quadrupole gradient	T/m	75.4
Arc beam screen aperture, H	mm	62.4
V	mm	47.4
RF voltages		
$h = 315$ injection & ramp	MV	0.6
$h = 630$ bunch splitting	MV	0.6
$h = 1260$ bunch splitting	MV	0.6
$h = 2520$ bunch compression	MV	6.0
$h = 7560$ bunch compression	MV	20.0

Table 4.2: Hadron Storage Ring lattice, layout, and RF parameters

4.0.3 Radial shift

Parameter	Unit	Values					
		protons				gold	
Total energy	GeV or GeV/u	41	100	133	275	40.7	110 .0
Relativistic γ		43.7	106.6	141.8	293.1	43.7	118.1
Circumference shift	mm	-908.7	-73.4	0.0	73.1	-908.7	-42.1
Average radial shift	mm	–	-11.7	0.0	11.6	–	-6.7

Table 4.3: Radial shift parameters in the Hadron Storage Ring

4.0.4 Transition crossing

Parameter	Unit	Value
Transition energy	GeV	22.94
Phase Advance per Arc Cell	°	87°, 85°
Transition γ		24.64
RF voltage	kV	200
RF harmonic number		315
Stable phase	rad	0.074
Acceleration rate	γ/s	0.5
Momentum spread max.		0.0055
Nonlinear factor α_1		-0.53×10^{-3}

Table 4.4: Transition crossing parameters in the Hadron Storage Ring

Chapter 5

Radio Frequency systems

5.0.1 RF summary parameters

RF System	Function	Location	Freq. MHz	Cavity type	Cavities per mod.	Voltage per cav. MV	Power kW
Electron accelerators							
Preinjector	Buncher		197	NCRF	1	0.4	
	Capture 1		1300	NC SW	1	5	
	Capture 2		1300	NC SW	1	10	
	Linac		2856	NC TW	14	60	
BAR	Store		52.88	NCRF	1	0.1	
RCS commissioning	Accelerate, store		197	NCRF, 1-cell		0.8	
RCS	Accelerate, store	IR4	591	SRF, 5-cell	8/8	17	
ESR	Store	IR10	591	SRF, 1-cell	18/9	3.5	800
LEC	Main	IR2	197	NCRF	17	0.85	
	Energy cornn.	IR2	591	NCRF	4	0.375	
	Deflecting	IR2	197	NCRF	1	0.15	
		IR2	24.6	NCRF	1	0.01	
Hadron Storage Ring							
	Capture/Accel	IR4	24.6	NCRF, QWR	4	0.6	120
	Bunch Split 1	IR4	49.2	NCRF, QWR	3	0.6	120
	Bunch Split 2	IR4	98.5	NCRF, QWR	4	0.6	120
	Store 1	IR4	197	NCRF, 1-cell	7	0.85	65
	Store 2	IR10	591	SRF, 1-cell	6/3	3.5	70
Crab cavities							
HSR		IR6	197	SRF, RFD	8/4		35
HSR		IR6	394	SRF, RFD	4/4		35
ESR		IR6	394	SRF, RFD	2/2		35

Table 5.1: Radio Frequency systems summary. All superconducting RF cavities are designed to operate at 2 K. The RCS cavities will initially operate at 4.5 K, which is sufficient to accelerate electrons to 10 GeV.

Chapter 6

New superconducting magnets

Electron and hadron beams collide with a crossing angle of 25 mrad at Interaction Point (IP6). There is a total of 27 new superconducting magnets in Interaction Region 6 (IR6), with a mixture of dipoles, quadruples, correctors, and solenoids. Multiple construction technologies are leveraged, including Direct Wind (DW) and cable. Cable magnets use a Rutherford style cable (Ruth.), while the Direct Wind magnets use a six-around-one conductor (6-R-1). Except for dipoles B2pF and B0pF, the new superconducting dipoles and quadrupoles share interconnected cryostats – four on the hadron-forward side, and three in the hadron-rear. The combined function B0pF magnet resides in a dedicated cryostat that includes Q0eF. Some magnets are tapered, with minimum and maximum coil and beampipe radii.

Spin rotator solenoids. Electron spin rotation is performed by solenoids, with help from neighboring dipoles. There are two sets of solenoids on each side of IP6 – a pair of long solenoids used at 18 GeV, and a pair of short solenoids used at 5 GeV. Both pairs are powered at intermediate energies.

6.0.1 Hadron Storage Ring dipoles

		B0pF	B0ApF	B1pF	B1ApF	B2pF
Construction		DW	DW	Cable	Cable	Cable
Coil radius	mm	328	TBD	150	TBD	60
Beampipe radius	mm	TBD	43	135	168	50
Lattice length	m	< 1.2	< 0.6	< 3	< 1.5	2.47
Coil length	m	1.2	0.6	3	1.5	2.85
Reference radius	mm	50	26	45	45	30
Field	T	-1.3	3.3	3.4	2.7	6.28
Layers		2	12	1	1	2
Operating current	kA	1.2	2	11.9	13.4	8.2
Quench margin	%	> 20	> 20	> 20	> 20	> 20
Ramp rate	T/s	-0.0065	0.0165	0.017	0.0135	0.031
Temperature	K	1.9	1.9	1.9	1.9	1.9
Strand diameter	mm	0.473	0.473	1.065	1.065	1.065
Conductor style		6-R-1	6-R-1	Ruth.	Ruth.	Ruth.

Table 6.1: Superconducting dipoles in the Hadron Storage Ring. The quench margins listed here are the minimum requirements – larger and more realistic values will be included as the magnet designs develop.

6.0.2 Hadron Storage Ring quadrupoles

		Q1ApF	Q1BpF	Q2pF	Q1ApR	Q1BpR	Q2pR
Construction		Cable	Cable	Cable	DW	DW	DW
Focus/Defocus		D	D	F	D	D	F
Gradient max.	T/m	89	56	41	78	78	26
Coil rad. min.	mm	71	93		28.35		
max.	mm				34.65		
Pipe rad. min.	mm	56	78	131	20.0	28	54
max.	mm				25.5		
Lattice length	m	<1.46	<1.61	<3.8	<1.8	<1.4	<4.5
Coil length	m	1.46	1.61	3.8	1.8	1.4	4.5
Ref. radius	mm	45	45	83	17	19	36
Layers		2	1	2	8	8	4
Operating current	kA	11	16	8.5	0.9	0.89	1.3
Quench margin	%	> 20	> 20	> 20	> 20	> 20	> 20
Ramp rate	T/ms	-0.44	-0.28	0.21	-0.39	-0.39	0.17
Temperature	K	1.9	1.9	1.9	4.5	4.5	4.5
Strand diam.	mm				0.33	0.33	0.473
Conductor		Ruth.	Ruth.	Ruth.	6-R-1	6-R-1	6-R-1

Table 6.2: Superconducting quadrupoles in the Hadron Storage Ring. The quench margins listed here are the minimum requirements – larger and more realistic values will be included as the magnet designs develop.

6.0.3 Electron Storage Ring quadrupoles

		Q0eF	Q1eF	Q1eR	Q2eR
Construction		DW	DW	DW	DW
Corrector Focus/Defocus		D	F	D	F
Gradient max.	T/m	14	6.3	14	14.1
Coil radius	mm				
Beampipe rad min.	mm	26	64.8	47.6	64.5
max.	mm			55.5	
Lattice length	m	<1.2	<1.61	<1.8	<1.4
Coil length	m	1.2	1.61	1.8	1.4
Reference radius	mm	17	43	TBD	20
Layers		4	2	4	2
Operating current	kA	0.08	1	1.8	1.8
Quench margin	%	> 20	> 20	> 20	> 20
Temperature	K	4.5	1.9	4.5	4.5
Strand diameter	mm	0.33	1.0	0.33	1.0
Conductor style			6-R-1		6-R-1

Table 6.3: Superconducting quadrupoles in the Electron Storage Ring. The quench margins listed here are the minimum requirements – larger and more realistic values will be included as the magnet designs develop. The abbreviation “DW” stands for Direct Wind.

6.0.4 Spin rotator solenoids

Parameter	Unit	Values	
		Short	Long
Coil radius	mm	86.0	86.0
Beampipe radius	mm	50.0	50.0
Lattice length	m	2.5	6.2
Coil length	m	1.874	5.68
Reference radius	mm	33.0	33.0
Integrated field	Tm	15.30	46.75
Layers		33	33
Operating current	kA	0.866	0.874
Quench margin	%	> 20	> 20
Inductance	mH	6.04	18.71
Stored energy	MJ	2.26	7.14
Temperature	K	1.9	1.9
Strand diameter	mm	1.6	1.6
Conductor style		WIC	WIC

Table 6.4: Superconducting spin rotator solenoids. The quench margins listed here are the minimum requirements – larger and more realistic values will be included as the magnet designs develop.

Chapter 7

Hadron Cooling

The Low Energy Cooler (LEC) uses electron cooling to decrease the vertical emittance of proton bunches just after they are injected into the Hadron Storage Ring. Electron cooling will work for any ions with the same relativistic γ as protons at injection in the HSR, with a cooling rate that scales like Z^2 / A . Cooling protons is the most demanding application.

Electron and proton beams co-move in a common straight section. A second harmonic RF system is used to alleviate space-charge effects during cooling, producing a longitudinally flattened proton bunch distribution with a peak current that is reduced by a factor of two. After cooling the proton bunches are accelerated to the collision energy, and are each split into four bunches. The LEC uses a 12.5 MeV RF electron accelerator to deliver electron bunches for hadron beam cooling.

7.0.1 Proton cooling at injection

Parameter	Unit	Value
Relativistic gamma	–	25.4
Proton energy	GeV	23.8
Electron kinetic energy	MeV	12.5
Length of cooling section	m	170
Proton repetition rate	MHz	24.6
Proton bunch length, RMS	m	1.0
Proton beta function in the cooling section	m	150
Electron beta function in the cooling section	m	150
Total electron charge per proton bunch	nC	3
Single electron bunch charge (3 electron bunches)	nC	1
Average electron current	mA	74
Electron emittance in cooling section, normalized RMS	μm	<1.5
Electron angular spread in cooling section, RMS	μrad	25
Electron energy spread in cooling section, RMS	–	4×10^{-4}
Electron bunch length in cooling section, RMS	mm	40

Table 7.1: Proton cooling at injection

7.0.2 Low Energy Cooler

Parameter	Unit	Value
DC gun voltage	kV	400
Final electron kinetic energy	MeV	12.5
Main RF bunch frequency	MHz	197
Number of 197MHz NCRF cavities		17
Voltage per 197MHz RF cavity	kV	850
Energy correction RF frequency	MHz	591
Number of 591MHz NCRF correction cavities		4
Voltage per 591MHz cavity	kV	375
Number of 591MHz deflecting cavities		1
Voltage of deflecting cavity	kV	150
Number of 24.6MHz cavities		1
Voltage of 24.6MHz cavity	kV	10
Bunch train repetition rate	MHz	24.6
Photocathode material		CsK ₂ Sb
Laser wavelength	nm	532
Electron bunch full length on cathode	ps	400
Average electron current	mA	74
Beam dump power	kW	925

Table 7.2: Low Energy Cooler parameters

Chapter 8

Cryogenic Systems

The Cryogenics Systems consist of the existing Central Plant located near sector 5, and Satellite Plants located near their associated loads. The Central Plant provides 4.5 K cooling to HSR components, interfaces with the Satellite Plants, and processes return gas from the Satellite Plants. Satellite Plants provide 1.92 K and 2 K cooling to components located in IR4, IR6, and IR10 by processing the 4.5 K helium down to 1.92 K for spin rotators, or 2.0 K for other SRF components. They also return helium gas back to the Central Plant.

8.0.1 Central plant capacity and loads

Parameter	Unit	Value
Refrigeration capacity at 4.5 K	kW	55
Temperature	K	4.5
Pressure	bar	16
Compressor flow at full capacity	g/s	4500
Existing distribution length (RHIC)	km	~ 4
Liquefaction load (4.5 to 300 K)	g/s	45
Re-liquefaction load (4.5 to 4.5 K)	g/s	288
Non-isothermal load (4.5 to 10 K)	g/s	282
Non-isothermal load (45 to 80 K)	g/s	102
Valve box count		8
Cryogenic controls I/O channel count		7500

Table 8.1: Cryogenics central plant capacity and loads, after configuration for EIC operations.

8.0.2 Detector

Parameter	Unit	Value
Temperature	K	4.6
Cryogenic load at 4.6 K	W	100
Flow rate at 4.6 K load	g/s	6
Load on thermal shield (45 to 80 K)	W	600
Flow rate on thermal shield	g/s	3
Current lead intercepts load (4.5 to 300 K)	W	850
Flow rate on current lead intercepts (4.5 to 80 K)	g/s	0.5
Electrical cryogenic heater count		4
Cryogenic distribution length	m	40
Valve box count		2
Cryogenic valve count		14

Table 8.2: ePIC Detector Solenoid cryogenics

8.0.3 Satellite plants

Parameter	Unit	Values		
		IR4	IR6	IR10
Cryogenic load at 2 K	W	500	650	1250
Temperature – main load	K	2	1.92	2
Temperature – SRF cryomodules	K	2	2	2
Flow rate at 2 K load	g/s	24	45	63
Current lead intercepts load (4.5 to 80 K)	W	TBD	6050	2220
Flow rate on current lead intercepts (4.5 to 80 K)	g/s	TBD	17.14	24
Cryogenic distribution length	m	TBD	168	163
Valve box count		8	19	12
Cryogenic valve count		56	95	60
I/O channel count		1272	1744	1220
Electrical cryogenic heater count		32	76	50

Table 8.3: Cryogenics satellite plants

Chapter 9

Detector

The EIC Detector, ePIC, consists of the central detector and several auxiliary detectors that are embedded in the hadron and electron accelerator lattice.

Due to the asymmetric electron ion collisions the ePIC detector consists of the central barrel detector and an electron and hadron endcap. Each region has an hadron, electromagnetic calorimeter, tracking and particle identification detectors. The ePIC detector has a 1.7 T solenoid with a 2 T stretch goal for the magnetic field. The auxiliary detectors are a zero-degree calorimeter, tracking detectors, and electromagnetic calorimetry embedded in the B0 magnet, as well as tracking detectors in the vacuum of the hadron beam to detect particles that scatter under small angles (Roman Pots) and charged particles from heavy ion decays (off-momentum detectors).

A luminosity detector is embedded in the electron lattice, with a tagger that detects electrons scattered under very small angles to reach Q^2 values below 0.1 GeV^2 .

9.0.1 Geometry and IR6

Parameter	Unit	Values		Comment
		Hadron	Lepton	
Machine element free region				
ePIC longitudinal	m	−4.5 to +5.0		
ePIC angular	deg.	1.5		
Hadron calorimeter back-face to first magnet front-face	m	0.5		
Central barrel radius	m	8.23		
Interaction point location				
Vertical, above floor	m	4.32		
H, inside current RHIC IP6	m	0.97		
BO detector angular range				As rotationally symmmetric as possible
minimum	mrاد	5.5		
maximum	mrاد	20.0		
Zero Degree Calorimeter				
Horizontal size	m	0.60		
Vertical size	m	0.60		
Longitudinal size	m	2.0		
Scattering angle acceptance				
ep collisions				All energies
Protons	GeV	$0.18 < p_t < 1.3$		Cone ≤ 5 mrad
Protons		$0.5 < x_L < 1.0$		$x_L = E'_p/E_{beam}$
Neutrons	GeV	$p_t < 1.3$		
eA collisions				Protons and neutrons
$\sqrt{s} = 50$ GeV	mrاد	$\Theta < 6.0$		
$\sqrt{s} = 100$ GeV	mrاد	$\Theta < 4.0$		
Bethe-Heitler photon acceptance	mrاد		± 1.0	
Luminosity variation $\delta L/L$			≤ 0.01	

Table 9.1: Detector geometry and interaction region parameters

9.0.2 Auxiliary detectors

Beam	Detector	Distance from IP6 m
Hadron	Roman Pot Station 1	33.56
	Station 2	35.06
	Off-momentum detectors Plane 1	25.45
	Plane 2	26.95
	Front Face ZDC	35.8
	Center pC - Polarimeter	-30.65
Electron	Luminosity Monitor exit window	-24.51
	Pair spectrometer Front Face	-63.79
	Low Q^2 Tagger 1	-27.36
	Tagger 2	-40.94
	Compton Polarimeter	
	Photon Detector	43.87
	Electron Detector	60.32
	Laser IP	75.37

Table 9.2: Auxiliary detector location measured from the interaction point. Positive coordinates are referencing the forward side and negative coordinates the rear side.

9.0.3 Solenoid parameters

Parameter	Unit	Value	Comment
Central field at interaction point	T	2.0	
Operating field range	T	0.5 – 2.0	
Field polarity		Bipolar	
Coil length	m	3.492	
Warm bore diameter	m	2.84	
Cryostat length	m	< 3.85	
Cryostat outer diameter	m	< 3.54	
Flat field area			
Longitudinal	m	± 1.0	
Radial	m	0.80	
Field uniformity	%	12.5	
RICH detector area			
Longitudinal	m	1.80 to 2.80	
Projectivity	mrad	0.1	At 30 GeV/c
Projectivity	T/(Amm ²)	10.0	
Stray field requirement	G	10.0	$z = -5.3 \text{ m}, +7.4 \text{ m}$
Charging voltage	V	10	
Fast discharge voltage max.	V	500	
Quench hot spot temperature	K	< 150	
Temperature margin	K	> 1.5	
Current margin	%	< 30	
Charging time	h	2 – 3	
Cooldown time	weeks	3 – 4	
Cooling scheme		Thermosiphon	
Operating temperature	k	4.5	

Table 9.3: ePIC detector solenoid parameters

Chapter 10

Infrastructure

The Electron-Ion Collider Conventional Facilities provide the buildings, utility systems and site infrastructure systems necessary to enable installation and operation of the EIC electron injector, electron and hadron storage rings, the interaction region, the ePIC detector, and support components.

The EIC project leverages the existing accelerator tunnel and related existing buildings that were constructed in the late 1970's for the Relativistic Heavy Ion Collider. The primary EIC facilities, typically reinforced concrete structures, are located adjacent to the existing Insertion Regions IR2, IR4, IR6, IR8, IR10, and IR12. Smaller buildings that house power supplies serving tunnel magnet systems are located outside the berm and tunnel profile, at sectors 1, 3, 5, 7, 9, and 11. These buildings also contain the mechanical cooling water pump and heat exchanger systems that serve magnet, power supply and vacuum beam pipe cooling.

The largest new EIC building will be located at IR10. This building (1010C) houses the RF equipment that powers the RF cavities located within the existing Insertion Region. This complex also includes water cooling and electrical power distribution systems for the loads in this area.

Building 638 is also expanded, adding two additional 69 kV/13.8 kV transformers and associated switchgear to distribute additional power to the EIC site via new overhead lines.